

Lab Project
EE 457 Fall 2024
Lab #1

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For: Prof. Thuc

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Purpose of Description

The purpose of this lab is to gain a good understanding about Fourier Series and Fourier Transform. And to learn how to implement Fourier analysis techniques using MATLAB/Simulink.

Design and Implementation Procedure

First, I create a rectangular pulse function in the time domain.

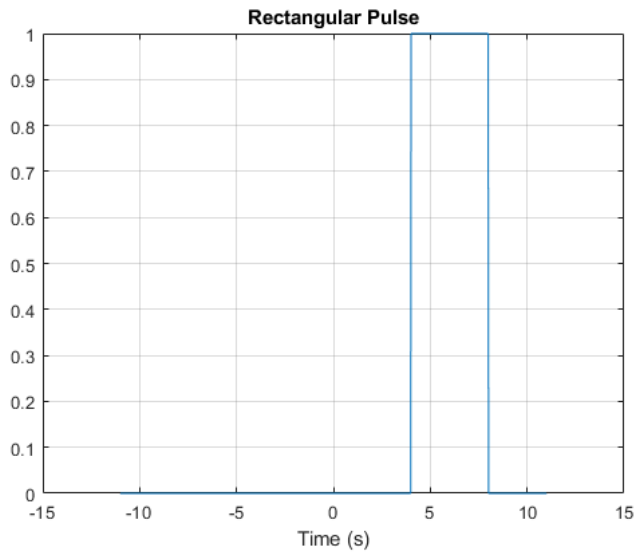


Figure 1: Rectangular pulse centered at 6 seconds with a pulse width of 4 seconds created with the rectangular pulse function.

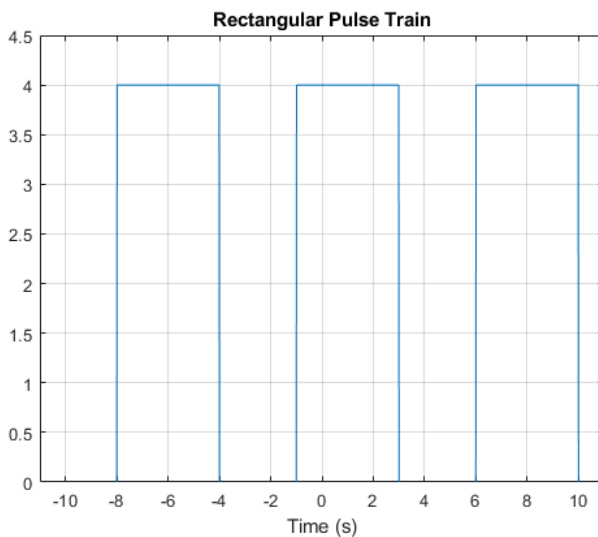


Figure 2: Rectangular pulse train starting at -8 seconds, with a period of 7 seconds and pulse width of 4 seconds.

Next, I found the Fourier Series in exponential form and trigonometric form of the pulse train in Figure 2.

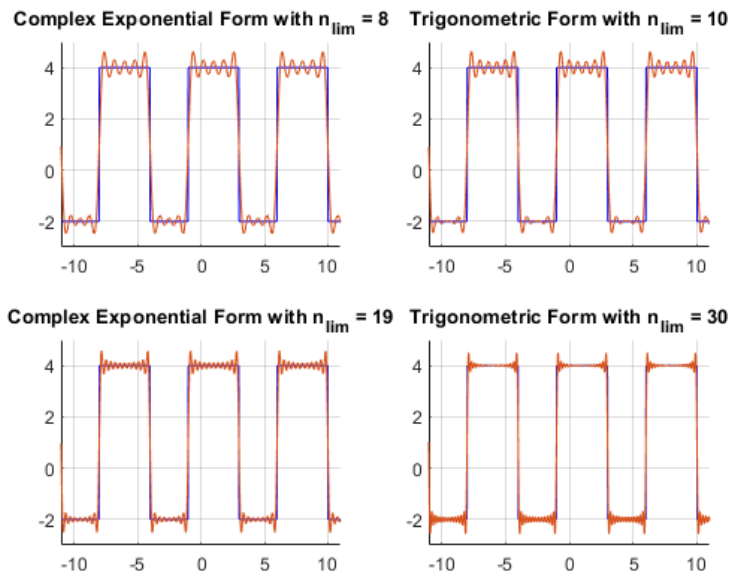


Figure 3: Fourier Series of a rectangular pulse train in trigonometric form and complex exponential form with varying limits of n .

Afterwards, I calculated the Fourier Transform of a rectangular pulse. Then, I applied the inverse Fourier Transform to retrieve the original signal.

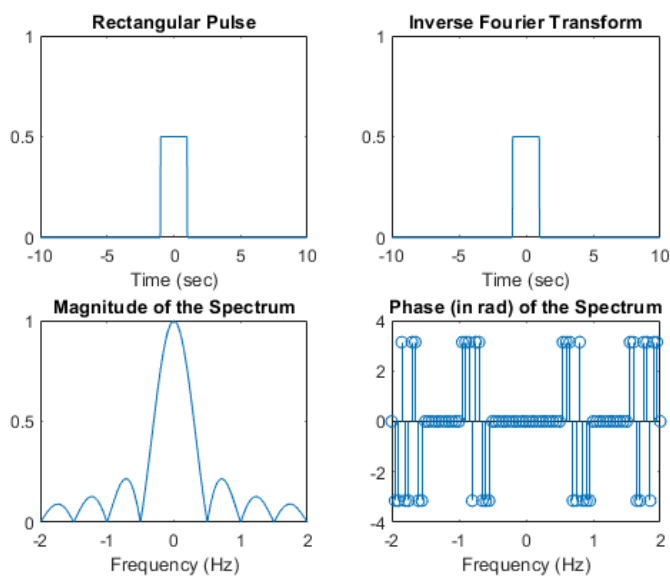


Figure 4: Fourier and Inverse Fourier Transform of a rectangular pulse plotting the magnitude and phase of the spectrum.

Lastly, I built a Simulink model to illustrate the Fourier Transform of a sinusoidal function.

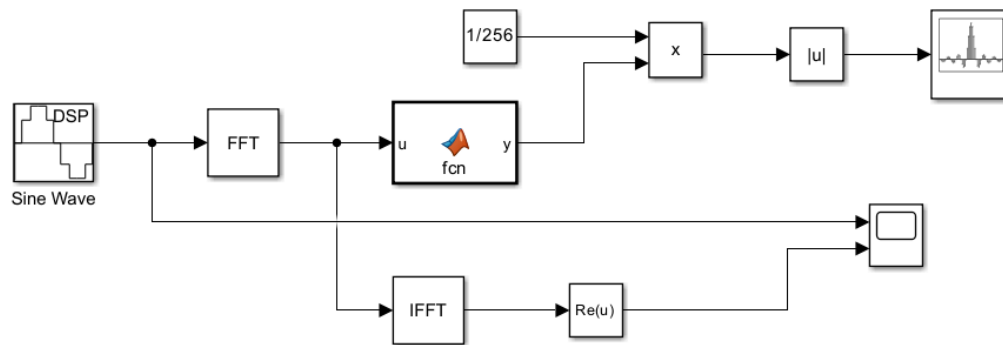


Figure 5: Simulink model of the Fourier Transform of a sinusoidal function.

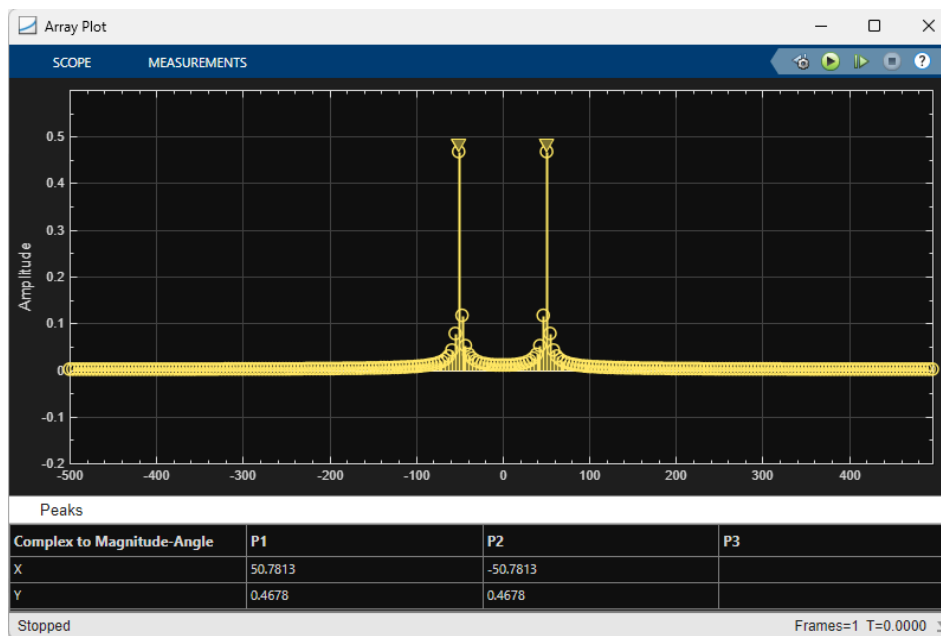


Figure 6: The Spectrum of the Fourier Transform of a sinusoidal function with a frequency of 50 Hz.

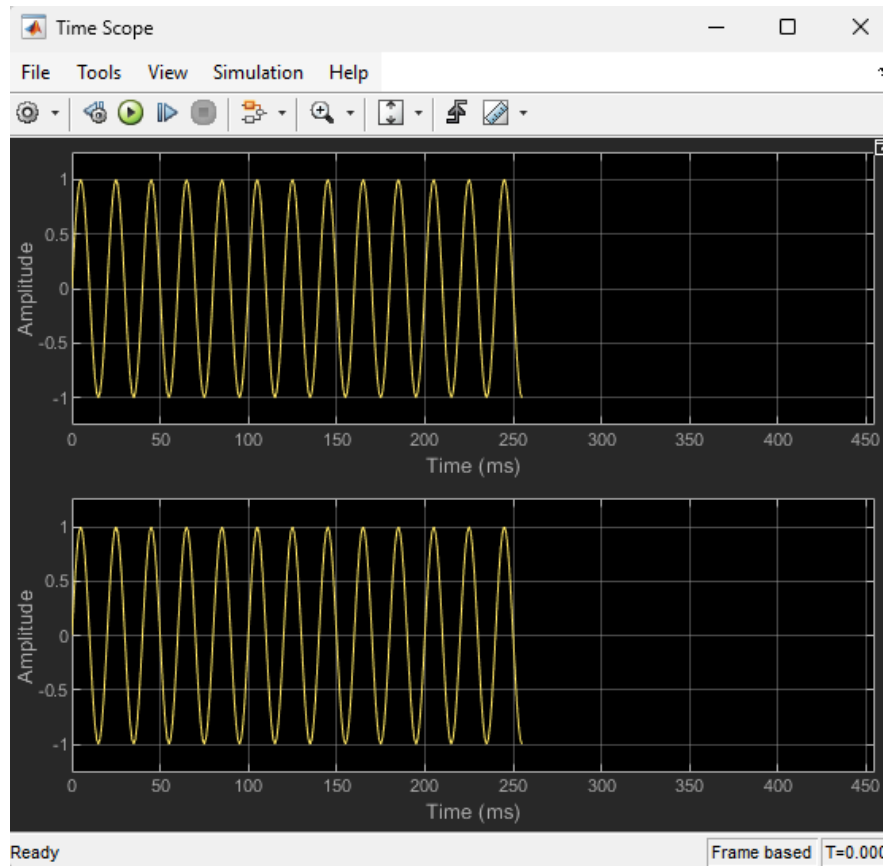


Figure 7: The time domain signal and inverse of the Fourier Transform of the same time domain signal.

Lessons Learned

Throughout this lab, I learned how to calculate the complex exponential and trigonometric Fourier Series of a rectangular pulse using MATLAB. I learned how to calculate the Fourier and Inverse Fourier Transform of a rectangular pulse in MATLAB and sinusoidal signal in Simulink. I learned that the Fourier Series is used to approximate a time signal.